

$\hat{\beta}_j(a)/\hat{\sigma}_{\beta_j}(a)$, where $\hat{\sigma}_{\beta_j}(a) = \sqrt{e_j' \hat{\Omega}_{\beta}(a) e_j} / \sqrt{N}$. Analogously, let

$$\hat{\Omega}_{\gamma} = \left(\frac{1}{N} \sum_i X_i X_i' \right)^{-1} \left(\frac{1}{N} \sum_i X_i X_i' \hat{u}_i^2 \right) \left(\frac{1}{N} \sum_i X_i X_i' \right)^{-1}$$

be the heteroskedasticity-robust variance estimator for $\hat{\gamma}$, the OLS estimate of γ , where $u_i = \mathbb{1}[Y_i > 0] - X_i' \hat{\gamma}$. The t -statistic for $\hat{\gamma}_j$ is then $\hat{t}_{\gamma_j} = \hat{\gamma}_j / \hat{\sigma}_{\gamma_j}$, where $\hat{\sigma}_{\gamma_j} = \sqrt{e_j' \hat{\Omega}_{\gamma} e_j} / \sqrt{N}$.

Proposition 7. *Suppose that $(\frac{1}{N} \sum_i X_i X_i')$ is full-rank and that $\hat{\sigma}_{\gamma_j} > 0$. If the function m satisfies parts 1 and 2 of [Proposition 1](#) and $\hat{\gamma}_j > 0$, then $\hat{t}_{\beta_j}(a) \rightarrow \hat{t}_{\gamma_j}$ as $a \rightarrow \infty$.*

It follows that when the units of Y are made large, the t -statistic for a treatment effect estimate for $m(Y)$ estimated using OLS will converge to the t -statistic for the OLS estimate of the extensive margin. [Appendix Figure 1](#) shows that, indeed, the t -statistics for estimates using $\text{arcsinh}(Y)$ in the *AER* tend to be close to the t -statistics for the extensive margin, and tend to become even closer after rescaling the units by a factor of 100.

Proof of Proposition 6. Fix $j \in \{2, \dots, J\}$. Note that $\beta(0) = E_Q[XX']^{-1} E[Xm(0)]$, is the coefficient from a regression of a constant outcome $m(0)$ on X , and thus $\beta_1(0) = m(0)$ while $\beta_k(0) = 0$ for $k \geq 2$. Thus $\beta_j(0) = 0$. To complete the proof, we will first show that $\beta_j(a_n) = \gamma_j \log(a_n) + o(\log(a_n))$. Hence, if $\gamma_j > 0$, then $|\beta_j(a)| \rightarrow \infty$ as $a \rightarrow \infty$. We will then establish that $\beta_j(a)$ is continuous for $a \in [0, \infty)$. The fact that one can obtain any positive value for $|\beta_j(a)|$ then follows from the intermediate value theorem.

For ease of notation, let $\nu' = e_j' E_Q[XX']^{-1}$, so that $\beta_j(a) = E_Q[\nu' X m(aY)]$.

We first show that $\beta_j(a_n) = \gamma_j \log(a_n) + o(\log(a_n))$. Consider a sequence $a_n \rightarrow \infty$, and assume without loss of generality that $a_n > e$. Let $f_n(x, y) = \nu' x \cdot m(a_n y) / \log(a_n)$. Observe that $f_n(x, y) \rightarrow \nu' x \cdot \mathbb{1}[y > 0]$ pointwise, since $f_n(x, 0) = \nu' x \cdot m(0) / \log(a_n) \rightarrow 0$, while for $y > 0$,

$$f_n(x, y) = \nu' x \cdot \frac{m(a_n y)}{\log(a_n)} = \nu' x \cdot \frac{m(a_n y)}{\log(a_n y)} \frac{\log(a_n) + \log(y)}{\log(a_n)} \rightarrow \nu' x,$$

where we use the fact that $m(y)/\log(y) \rightarrow 1$ as $y \rightarrow \infty$. We showed in the proof to [Proposition 4](#) that

$$\frac{|m(a_n y)|}{\log(a_n)} \leq \kappa + 2 \cdot \mathbb{1}[y > 0] \cdot (1 + |\log(y)|),$$

which implies that

$$|f_n(x, y)| \leq |\nu' x \cdot (\kappa + 2 \cdot \mathbb{1}[y > 0] \cdot (1 + |\log(y)|))| =: \bar{f}(x, y).$$

Moreover, part 2 of the proposition implies that $\bar{f}(X, Y)$ is integrable. From the dominated convergence theorem, it follows that

$$\frac{\beta_j(a_n)}{\log(a_n)} = E_Q[f_n(X, Y)] \rightarrow E_Q[\nu' X \mathbb{1}[Y > 0]] = \gamma_j.$$

Hence, we see that $\beta_j(a_n) = \gamma_j \log(a_n) + o(\log(a_n))$. It follows that $|\beta_j(a_n)| \rightarrow \infty$ when $\gamma_j \neq 0$.

To complete the proof, we show continuity of $\beta_j(a)$. Fix $a \in [0, \infty)$, and consider a sequence $a_n \rightarrow a$. Assume without loss of generality that $a_n < a + 1$ for all n . Let $f_n(x, y) = \nu'x \cdot m(a_n y)$. From the continuity of m , we have that $f_n(x, y) \rightarrow \nu'x \cdot m(ay)$ pointwise. We showed in the proof to [Proposition 1](#) that there exists some κ (not depending on n) such that

$$|m(a_n y)| \leq \kappa + 2\mathbb{1}[y > 0] \cdot |\log(y)|.$$

Hence,

$$|f_n(x, y)| \leq |\nu'x \cdot (\kappa + 2\mathbb{1}[y > 0]|\log(y)|)|.$$

Moreover, the bounding function is integrable over the distribution of (X, Y) by part 2 of the proposition. Applying the dominated convergence theorem again, we obtain that

$$\beta_j(a_n) = E_Q[f_n(X, Y)] \rightarrow E_Q[\nu'X \cdot m(aY)] = \beta_j(a),$$

as needed. $\square$

Proof of Proposition 7. Consider $a_n \rightarrow \infty$. Applying [Proposition 6](#) to the empirical distribution, we have that $\hat{\beta}(a_n)/\log(a_n) = \hat{\gamma} + o(1)$. It follows that

$$\frac{1}{\log(a_n)}\hat{\epsilon}_i(a_n) = \frac{m(a_n Y_i)}{\log(a_n)} - \frac{\hat{\beta}(a_n)'X_i}{\log(a_n)} = \mathbb{1}[Y_i > 0] - \hat{\gamma}'X_i + o(1) = \hat{u}_i + o(1).$$

Since $\hat{\Omega}_n(a_n)$ is a continuous function of the $\hat{\epsilon}_i(a_n)^2$, we obtain that $\log(a_n)^{-2}\hat{\Omega}_\beta(a_n) \rightarrow \hat{\Omega}_\gamma$, and thus that $\log(a_n)^{-1}\hat{\sigma}_{\beta_j}(a_n) = \hat{\sigma}_{\gamma_j} + o(1)$. It follows that

$$\hat{t}_{\beta_j}(a_n) = \frac{\hat{\beta}_j(a_n)/\log(a_n)}{\hat{\sigma}_{\beta_j}(a_n)/\log(a_n)} = \frac{\hat{\gamma}_j + o(1)}{\hat{\sigma}_{\gamma_j} + o(1)} \rightarrow \frac{\hat{\gamma}_j}{\hat{\sigma}_{\gamma_j}} = \hat{t}_{\gamma_j},$$

as needed. $\square$

C Connection to structural equations models

Previous work has considered a variety of estimators for settings with zero-valued outcomes beginning with structural equations models rather than the potential outcomes model that we consider. These papers have reached different results, with some concluding that regressions with $\text{arcsinh}(Y)$ have the interpretation of an elasticity, and others showing that they are inconsistent and advocating for other methods (e.g. Poisson regression) instead. In this section, we interpret the results in those papers from the perspective of the potential outcomes model, and show that these diverging conclusions stem from different implicit assumptions about the potential outcomes, as well as a focus on different causal parameters.

Before discussing specific papers, we first note that, broadly speaking, structural equation models

can be viewed as constraining the joint distribution of potential outcomes. Observe that, for any pair of potential outcomes $(Y(1), Y(0))$, we can represent them as $(Y(1, U), Y(0, U))$ for some function $Y(d, u)$ and individual-level unobservable (or “structural error”) U . The potential outcomes framework we work with in this paper does not impose any functional form assumptions on $Y(d, u)$. Structural equation models, on the other hand, tend to specify explicit functional forms for $Y(d, u)$. In what follows, we consider the implicit restrictions placed on the potential outcomes as well as the target estimand in work related work that starts with a structural equations model.

C.1 Bellemare and Wichman (2020) and Thakral and Tô (2023)

Bellemare and Wichman (2020) consider OLS regressions of the form⁵¹

$$\operatorname{arcsinh}(Y) = \beta_0 + D\beta_1 + U. \quad (\text{C.1})$$

Note that when D is binary and randomly assigned, $D \perp\!\!\!\perp (Y(1), Y(0))$, then from the perspective of the potential outcomes model, the population coefficient β_1 is the ATE for $\operatorname{arcsinh}(Y)$. Bellemare and Wichman (2020) instead consider the interpretation of β_1 when (C.1) is treated as structural, i.e. if there are constant treatment effects of D on $\operatorname{arcsinh}(Y)$. From the perspective of the potential outcomes model, this amounts to imposing that the potential outcomes $Y(d) := Y(d, U)$ take the form

$$\operatorname{arcsinh}(Y(d, U)) = \beta_0 + d\beta_1 + U, \quad (\text{C.2})$$

where the individual-level random variable U takes the same value for all values of d . Under (C.2), we have that

$$\beta_1 = \operatorname{arcsinh}(Y(1, U)) - \operatorname{arcsinh}(Y(0, U)).$$

Since $\operatorname{arcsinh}(y) \approx \log(2y)$ for y large, it follows that $\beta_1 \approx \log(Y(1, U)/Y(0, U))$ when $Y(1, U)$ and $Y(0, U)$ are large. Thus, Bellemare and Wichman (2020) argue that β_1 approximates the semi-elasticity of the outcome with respect to d when the outcome is large. They likewise provide similar results for the elasticity of $Y(d, U)$ with respect to treatment when treatment is continuous. Their results thus imply that the ATE for $\operatorname{arcsinh}(Y)$ has a sensible interpretation as a (semi-)elasticity when the structural equation for the potential outcomes given in (C.2) holds.

It is worth emphasizing, however, that (C.2) will generally be incompatible with the data when both $Y(1)$ and $Y(0)$ have point-mass at zero, and $\beta_1 \neq 0$. Specifically, note that (C.2) implies that for all values of U ,

$$\operatorname{arcsinh}(Y(1, U)) - \operatorname{arcsinh}(Y(0, U)) = \beta_1.$$

If $\beta_1 > 0$, for example, this implies that $\operatorname{arcsinh}(Y(1, U)) > \operatorname{arcsinh}(Y(0, U))$, and hence $Y(1, U) > Y(0, U)$, since the $\operatorname{arcsinh}(y)$ function is strictly increasing for $y \geq 0$. However, since $Y(0, U) \geq 0$ by assumption, this implies that $Y(1, U) > 0$ with probability 1. Thus, the model in (C.2) is

⁵¹They also consider specifications with additional covariates on the right-hand side, although we abstract away from this for expositional simplicity.

incompatible with $P(Y(1) = 0) > 0$ if $\beta_1 > 0$. By similar logic, the model is also incompatible with $P(Y(0) = 0) > 0$ if $\beta_1 < 0$. In settings where there is point-mass at zero, the model that [Bellemare and Wichman \(2020\)](#) show gives β_1 an interpretation as a semi-elasticity will therefore typically be rejected by the data. It is also worth noting that even if there are no zeros in the data, the model in (C.2) will generally be sensitive to units, in the sense that if (C.2) holds for Y measured in dollars, it will generally not hold when Y is measured in cents. The validity of the interpretation of β_1 as an elasticity thus depends on having chosen the “correct” scaling of the outcome such that (C.2) holds.

Similar issues apply if we consider alternative transformations on the left-hand side of (C.1). For example, [Thakral and Tô \(2023\)](#) consider versions of (C.1) that replaces $\text{arcsinh}(Y)$ with the power function Y^k . They then consider the implied “semi-elasticities” of the form $\eta(y_0) = \beta_1/(ky_0^k)$. The parameter $\eta(y_0)$ has the interpretation as a structural semi-elasticity when d has a constant effect on Y^k . Specifically, if D is continuous and the structural equation

$$Y(d, U)^k = \beta_0 + d\beta_1 + U, \quad (\text{C.3})$$

holds, then $\eta(y_0) = (\frac{\partial}{\partial d}Y(d, U))/Y(d, U)$ evaluated at $Y(d, U) = y_0$, so $\eta(y_0)$ corresponds to the semi-elasticity of $Y(d, U)$ with respect to d . However, as with (C.2), (C.3) is generally incompatible with settings in which $P(Y(d, U) = 0)$ for multiple values of d . For example, if $\beta_1 > 0$, then $Y(0, U) \geq 0$ implies that $Y(1, U) > 0$. Equation (C.3), which gives a causal interpretation to $\eta(\beta_0)$ as a semi-elasticity, will thus generally be incompatible with settings in which some units have $Y = 0$ under multiple treatment statuses.

C.2 [Cohn et al. \(2022\)](#)

[Cohn et al. \(2022\)](#) consider structural equations of the form

$$Y = \exp(\alpha + D\beta)U. \quad (\text{C.4})$$

When $E[U | D] = 1$, they show that Poisson regression is consistent for β , whereas regressions of $\log(1 + Y)$ or $\log(Y)$ on D may be inconsistent for β .⁵² Although [Cohn et al. \(2022\)](#) do not consider a potential outcomes interpretation of β , we can give β a causal interpretation if we impose that the potential outcomes take the form

$$Y(d, U) = \exp(\alpha + d\beta)U(d), \quad (\text{C.5})$$

where $E[U(d)] = 1$. Under (C.5), it follows that $\exp(\beta) = E[Y(1)]/E[Y(0)]$, i.e. the parameter

⁵²We thank Kirill Borusyak for an insightful discussion on this topic. Relatedly, in an influential paper, [Santos Silva and Tenreiro \(2006\)](#) consider the structural equations model $Y_i = \exp(X_i'\beta)U_i$ where $E[U_i | X_i] = 1$, and show that Poisson regression consistently estimates β while a regression using log on the left-hand side does not, although they do not provide any formal results on log-like transformations.